Product quality & reliability · Engxtech · data cut-off 30 Sep 2026

Fictional company, synthetic data.

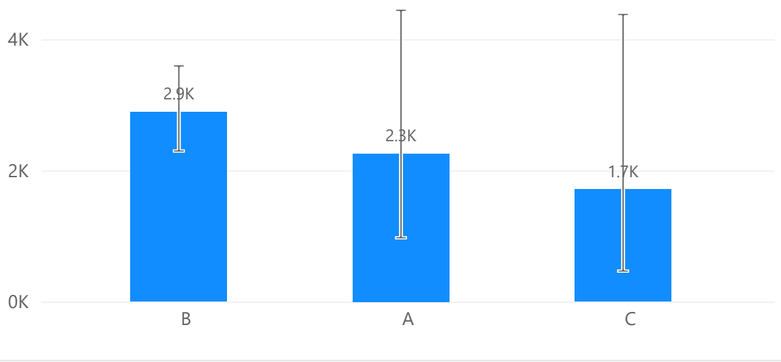
PartID

All

Validation test pass rate	Warranty claims	Warranty cost (USD)	Seat ring wear cost (USD)	DFMEA RPN reduction	High severity, no action
90.9%	234	94,776	88,246	46.4%	2

Seat-ring-wear ppm by seat ring rev (matured months, 95 % CI)

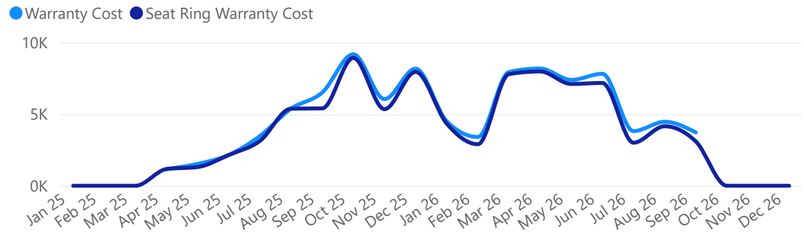
Field ppm, Wear Failures 90d and Matured Units by SeatRingRev



ppm = seat-ring-wear claims within 90 days of the build month, per million product units built; matured build months only (Mar 2025 to Jun 2026). Rev C: 4 failures, 1 matured month: an early signal.

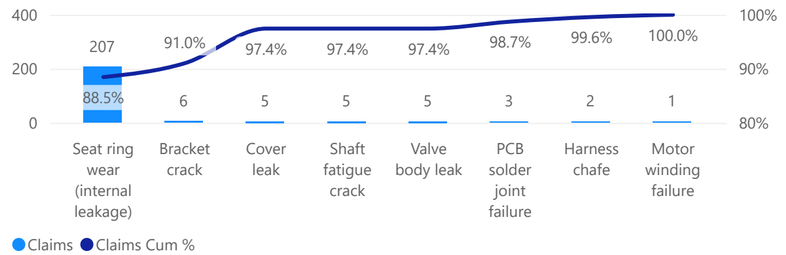
Warranty cost by failure month (USD)

Warranty Cost and Seat Ring Warranty Cost by MonthStart



Claims by failure mode (Pareto)

Claims and Claims Cum % by FailureMode



DFMEA risk: red = severity 7 or more, no action

LineID	PartID	FailureMode	S	RPN Before	RPN After	Action
DF-01	P300	Seat ring wear (internal leakage)	8	280	72	Tighter OD tolerance, rev C
DF-08	E700	Harness chafe on bracket	9	216	72	Grommet added, rev B
DF-09	E800	Solder joint failure	7	210	84	Thermal profile change, rev C
DF-04	P200	Valve body leak at flange joint	7	140	140	
DF-02	P300	Seat ring migration in body bore	6	120	72	Interference fit check added
DF-07	E600	Motor winding failure	8	120	120	
DF-03	P100	Shaft fatigue crack	9	108	54	Fillet radius increased, rev B